DATA ANALYTICS

Major PROJECT - 1

END-TO-END HORTICULTURE ANALYTICS:
CROP TYPE, CROP NAME, MANDAL, DISTRICT,
AREA AND PRODUCTION

Collecting Raw Data

Data Sources

- Collected raw data from government sources with real-time data access
- Dataset spans from 2020 to 2023 containing lakhs of records
- Initial loading time was extensive due to massive data volume

Sampling Strategy

- Random selection from 3-year dataset to reduce data size
- Thereby enabling efficient processing and analysis
- Balanced representation across all time periods

Data Structure

year	district	division	mandal	crop_type
crop_name	season	area_production	value	note

Random Selection Methodology

■ Process Flow

Raw Dataset	→	2020-2023 Complete Data
Random Filter	→	Statistical Sampling
Subset Creation	→	Manageable Sample Size
Data Merge	→	Combined Working Dataset

■ Selection Criteria

- **Random Distribution**
  Ensures unbiased representation across all districts and time periods
- **Proportional Coverage**
  Maintains relative proportions of crop types and geographical regions
- **Temporal Balance**
  Equal representation from each year (2020, 2021, 2022, 2023)

★ Random sampling ensures statistical validity while reducing computational load

Random Selection Methodology

Step-by-Step (Power Query)

Step 1: Load data

- Excel → **Data → Get Data → From Table/CSV**
- Open **Power Query Editor**

Step 2: Add Random Column

1. Go to **Add Column → Custom Column**
2. Formula:

= Number.Random()

3. Name it: Random_No

Step 3: Sort by Random

- Click Random_No
- **Sort Ascending**

This shuffles your entire dataset randomly

Step 4: (Important) Grouping Logic

Randomness **based on year, district, crop_type, crop_name**

Go to **Home → Group By(Select advanced)**

- Group By columns:
  - year
  - district
  - crop_type
  - crop_name
- New column name: AllData
- Operation: **All Rows**

Step 5: Pick Random Rows Per Group

Add **Custom Column**:

= Table.FirstN([AllData], 1000)

Picks **1000 random rows per group**

Step 6: Expand Table

- Click expand icon on the new column
- Remove Random_No

Step 7: Final Count Control

After expansion:

Home->Keep rows

- Use **Keep Top Rows → 1000**

Data Cleaning Process

■ Column Removal

division
Not required - district and mandal sufficient

season
Inaccurate data - mode would misclassify (e.g., mango as summer fruit)

■ Value Cleaning

Zero Values
Removed - indicate no production/area

Blank Entries
Removed from crop_type and values columns

Typos Fixed
"arearea" → "area" (find & replace)

■ Note Column Fix

For mixed crops with missing note data, implemented SWITCH case:

```
=SWITCH([@area_production],
"Area","Area in Acres",
"Production","Production in Metric Tonnes",
"")
```

→ Data quality improvements enable accurate analytical insights

Before & After Cleaning

BEFORE Cleaning

Data Issues

- Inconsistent column entries
- Missing values in critical fields
- Duplicate or irrelevant columns
- Typographical errors

Impact

- Reduced analytical accuracy
- Biased statistical results
- Incomplete insights

AFTER Cleaning

Improvements

- Standardized data structure
- Complete value entries
- Relevant columns only
- Corrected terminology

Benefits

- Enhanced data reliability
- Accurate trend analysis
- Actionable insights

● Quality: POOR → EXCELLENT ● Completeness: PARTIAL → FULL ● Usability: LIMITED → HIGH

Before Cleaning

year	district	division	mandal	crop_type	crop_name	season	area_production	value	note
2020-21	Karimnagar	Choppadandi	Ramadugu	Perennial Fruit Crops	Mango		Area	661	Area in Acres
2020-21	Karimnagar	Huzurabad	Ellandakunta	Perennial Fruit Crops	Mango		Area	40.03	Area in Acres
2020-21	Karimnagar	Huzurabad	Huzurabad	Perennial Fruit Crops	Mango		Area	111.32	Area in Acres
2020-21	Karimnagar	Huzurabad	Jammikunta	Perennial Fruit Crops	Mango		Area	10.02	Area in Acres
2020-21	Karimnagar	Huzurabad	V.Saidapur	Perennial Fruit Crops	Mango		Area	155.3	Area in Acres
2020-21	Karimnagar	Huzurabad	Veenavanka	Perennial Fruit Crops	Mango		Area	169.01	Area in Acres
2020-21	Karimnagar	Karimnagar	Karimnagar	Perennial Fruit Crops	Mango		Area	0	Area in Acres
2020-21	Karimnagar	Karimnagar	Karimnagar Rural	Perennial Fruit Crops	Mango		Area	169.37	Area in Acres
2020-21	Karimnagar	Karimnagar	Kothapally	Perennial Fruit Crops	Mango		Area	145.13	Area in Acres
2020-21	Karimnagar	Manakondur	Chigurumamidi	Perennial Fruit Crops	Mango		Area	892.16	Area in Acres
2020-21	Karimnagar	Manakondur	Ganneruvaram	Perennial Fruit Crops	Mango		Area	336.26	Area in Acres
2020-21	Karimnagar	Manakondur	Manakondur	Perennial Fruit Crops	Mango		Area	1201.07	Area in Acres
2020-21	Karimnagar	Manakondur	Shankarapatnam	Perennial Fruit Crops	Mango		Area	103.07	Area in Acres
2020-21	Karimnagar	Choppadandi	Choppadandi	Perennial Fruit Crops	Mango		Production	2873.97	Production in Metric Tonnes
2020-21	Karimnagar	Choppadandi	Gangadhara	Perennial Fruit Crops	Mango		Production	3500.82	Production in Metric Tonnes
2020-21	Karimnagar	Choppadandi	Ramadugu	Perennial Fruit Crops	Mango		Production	2584.51	Production in Metric Tonnes
2020-21	Karimnagar	Huzurabad	Ellandakunta	Perennial Fruit Crops	Mango		Production	156.52	Production in Metric Tonnes
2020-21	Karimnagar	Huzurabad	Huzurabad	Perennial Fruit Crops	Mango		Production	435.26	Production in Metric Tonnes
2020-21	Karimnagar	Huzurabad	Jammikunta	Perennial Fruit Crops	Mango		Production	39.18	Production in Metric Tonnes
2020-21	Karimnagar	Huzurabad	V.Saidapur	Perennial Fruit Crops	Mango		Production	607.22	Production in Metric Tonnes
2020-21	Karimnagar	Huzurabad	Veenavanka	Perennial Fruit Crops	Mango		Production	660.83	Production in Metric Tonnes
2020-21	Karimnagar	Karimnagar	Karimnagar	Perennial Fruit Crops	Mango		Production	0	Production in Metric Tonnes
2020-21	Karimnagar	Karimnagar	Karimnagar Rural	Perennial Fruit Crops	Mango		Production	662.24	Production in Metric Tonnes
2020-21	Karimnagar	Karimnagar	Kothapally	Perennial Fruit Crops	Mango		Production	567.46	Production in Metric Tonnes
2020-21	Karimnagar	Manakondur	Chigurumamidi	Perennial Fruit Crops	Mango		Production	3488.35	Production in Metric Tonnes

After Cleaning

year	district	mandal	crop_type	crop_name	area_production	value	note
2020-21	Karimnagar	Choppadandi	Perennial Fruit Crops	Mango	Area	735.03	Area in Acres
2020-21	Karimnagar	Shankarapatnam	Perennial Fruit Crops	Mango	Area	103.07	Area in Acres
2020-21	Karimnagar	Manakondur	Perennial Fruit Crops	Mango	Area	1201.07	Area in Acres
2020-21	Karimnagar	Ganneruvaram	Perennial Fruit Crops	Mango	Area	336.26	Area in Acres
2020-21	Karimnagar	Chigurumamidi	Perennial Fruit Crops	Mango	Area	892.16	Area in Acres
2020-21	Karimnagar	Kothapally	Perennial Fruit Crops	Mango	Area	145.13	Area in Acres
2020-21	Karimnagar	Veenavanka	Perennial Fruit Crops	Mango	Area	169.01	Area in Acres
2020-21	Karimnagar	V.Saidapur	Perennial Fruit Crops	Mango	Area	155.3	Area in Acres
2020-21	Karimnagar	Jammikunta	Perennial Fruit Crops	Mango	Area	10.02	Area in Acres
2020-21	Karimnagar	Huzurabad	Perennial Fruit Crops	Mango	Area	111.32	Area in Acres
2020-21	Karimnagar	Ellandakunta	Perennial Fruit Crops	Mango	Area	40.03	Area in Acres
2020-21	Karimnagar	Ramadugu	Perennial Fruit Crops	Mango	Area	661	Area in Acres
2020-21	Karimnagar	Gangadhara	Perennial Fruit Crops	Mango	Area	895.35	Area in Acres
2020-21	Karimnagar	Karimnagar Rural	Perennial Fruit Crops	Mango	Area	169.37	Area in Acres
2020-21	Kothagudem	Palwancha	Spices	Turmeric	Area	0.05	Area in Acres
2020-21	Kothagudem	Palwancha	Spices	Turmeric	Area	140.31	Area in Acres
2020-21	Kothagudem	Julurpad	Spices	Turmeric	Area	3	Area in Acres
2020-21	Kothagudem	Aswapuram	Spices	Turmeric	Area	14.8	Area in Acres
2020-21	Kothagudem	Burgampadu	Spices	Turmeric	Area	3	Area in Acres
2020-21	Kothagudem	Yellandu	Spices	Turmeric	Area	1.05	Area in Acres
2020-21	Kothagudem	Tekulapalle	Spices	Turmeric	Area	0.39	Area in Acres

Insights: Yearly Trends & Geographical Analysis

■ Yearly Trends

- **Cultivated Area Changes**
  How has total horticulture cultivated area changed over different years?
- **Production Variations**
  How has total horticulture production varied year-wise?
- **Growth Proportionality**
  Is production growth proportional to cultivated area growth?
- **Average Production**
  What is the average production value observed over the years?

■ Geographical Insights

- **District Contributions**
  Which districts contribute highest total area and production?
- **Area-Production Gap**
  Districts with high cultivated area but low production?
- **Mandal Specialization**
  Which mandals contribute disproportionately to area/production?
- **Crop Diversity**
  Which districts/mandals have highest crop diversity?

★ These analytical questions guide the entire data exploration process

Insights: Crop Analysis & Data Quality

■ Crop & Type Analysis

- **Area Distribution**
  How is cultivated area distributed across crop types?
- **Production Leaders**
  Which crop types contribute most to total production?
- **Individual Crop Performance**
  Which crops occupy largest area and highest production?
- **Expansion Trends**
  Which crop types are expanding in cultivated area over time?
- **Single-District Crops**
  Are there crops reported only in a single district?

■ Data Quality Assessment

- **Reporting Patterns**
  What proportion of records report only area vs. only production?
- **Production Variability**
  Which districts show highest variability (standard deviation) in production?
- **Average by Type**
  What is the average production per record by crop type?
- **Data Completeness**
  Assessment of missing values and data gaps

→ Comprehensive analysis covering crop performance and data integrity

SQL Database Structure

■ Database: agriculture

year table
Columns: year_id, year

district table
Columns: district_id, district

mandal table
Columns: mandal_id, mandal, district_id

crop table
Columns: crop_id, crop_type, crop_name

■ Main Transaction Table

area_production table

- ap_id (Primary Key)
- year_id (Foreign Key)
- district_id (Foreign Key)
- mandal_id (Foreign Key)
- crop_id (Foreign Key)
- area_production
- value
- note

Data Import Method:
Table Data Import Wizard → Select respective table → Import cleaned data

★ Normalized structure enables efficient querying and data integrity

Python-SQL Connection

■ Library Imports

```
import mysql.connector
from mysql.connector import Error
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

- mysql.connector: Database connection
- Error handling: Prevents connection failures
- numpy: Numerical operations
- pandas: Data manipulation
- seaborn/matplotlib: Visualization

■ Connection Setup

Database Configuration:

```
db_config = {
"host": "localhost",
"user": "root",
"password": "2855@Mysql",
"database": "agriculture"
}
```

Establish Connection:

```
connection = mysql.connector
.connect(**db_config)
```

Create Cursor (Temporary Storage):

```
cursor = connection.cursor()
```

Retrieved data from SQL is stored temporarily in cursor

→ Seamless integration between SQL database and Python analytics environment

Data Visualization: Python Implementation

■ Core Libraries

- **pandas (pd)**
  Converts SQL query results into DataFrame format for easy manipulation

- **matplotlib.pyplot (plt)**
  Primary visualization library for creating static, animated, and interactive plots

- **seaborn (sns)**
  Statistical data visualization with attractive default styles

■ Visualization Strategy

- **Selective Insights**
  Only key insights visualized that demonstrate main analytical findings

- **Chart Types Used**
  - Bar Charts
  - Pie Charts
  - Scatter Plots
  - Line Graphs

- **Data Flow**
  SQL Query → Cursor → DataFrame → Visualization

★ Effective visualization transforms raw data into actionable insights

Visualization Examples

■ Chart Type 1: Bar Chart

District-wise Production
Comparative analysis of total production across districts

Purpose:
Compare categorical data (districts) against numerical values (production)

■ Chart Type 2: Pie Chart

Crop Type Distribution
Percentage share of different crop types

Purpose:
Show proportional composition of whole (crop type distribution)

■ Chart Type 3: Line Chart

Yearly Trends
Production/Area changes over time (2020-2023)

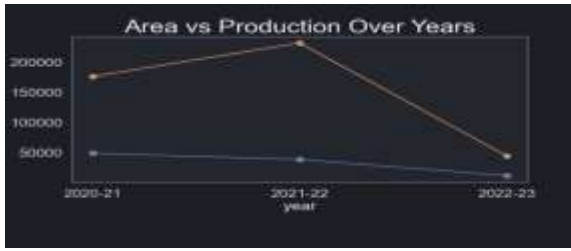
Purpose:
Display trends and patterns over continuous time periods

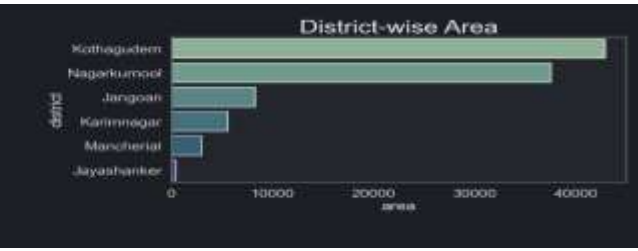
■ Chart Type 4: Scatter Plot

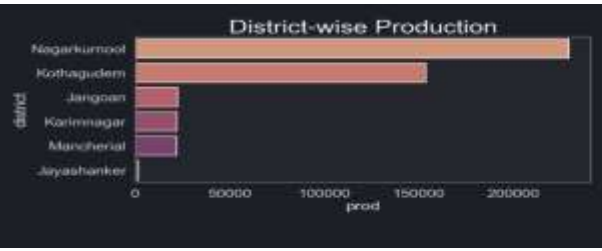
Area vs Production
Correlation between cultivated area and production

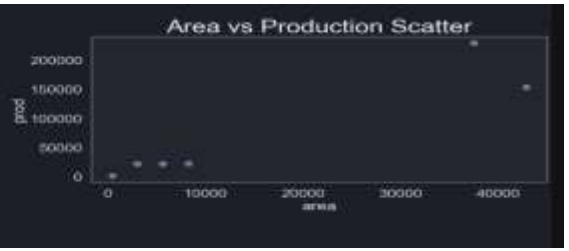
Purpose:
Identify relationships and correlations between two variables

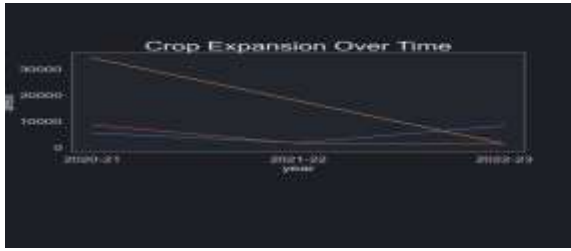
Data visualization: using python

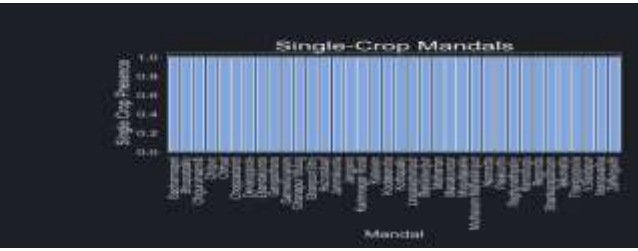


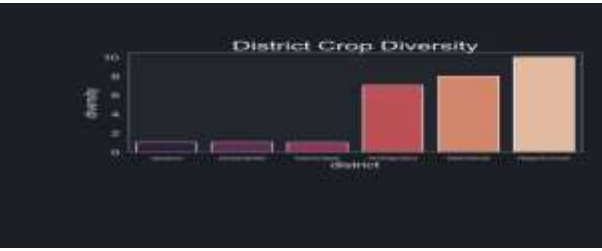


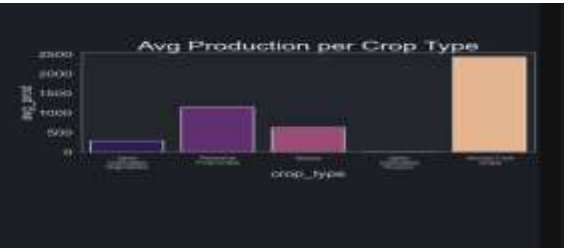


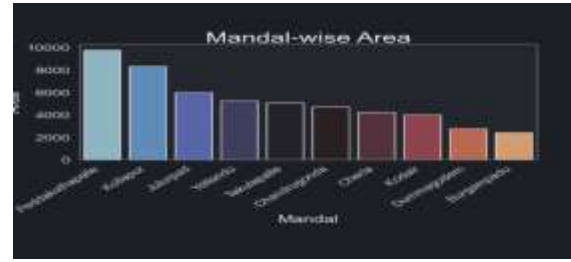


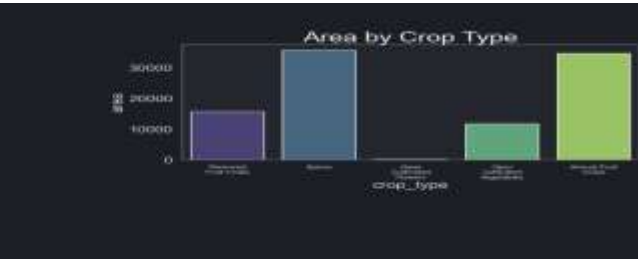


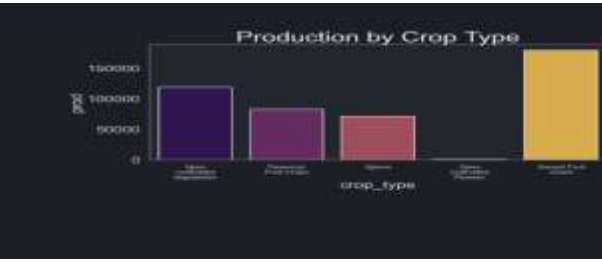


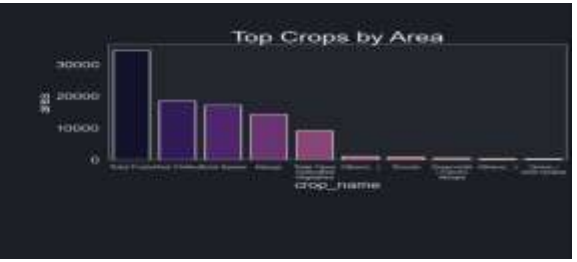


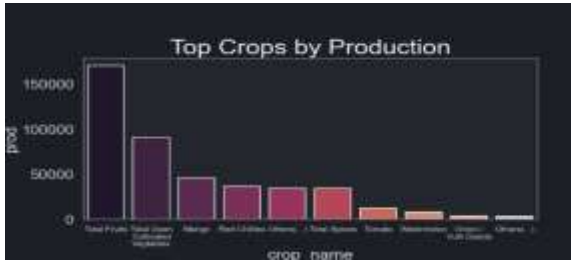


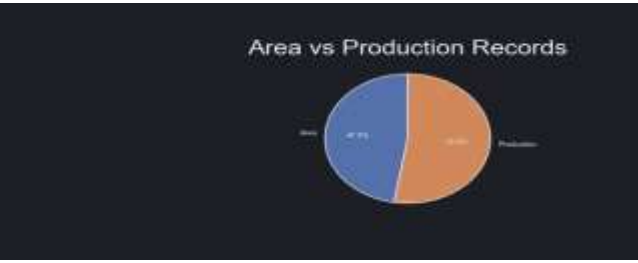


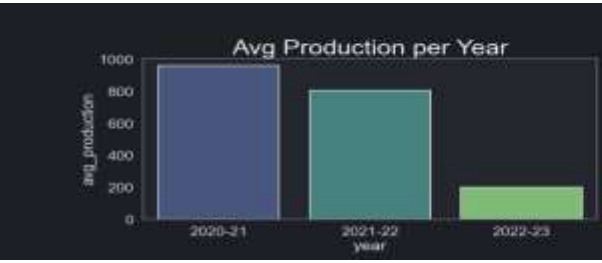


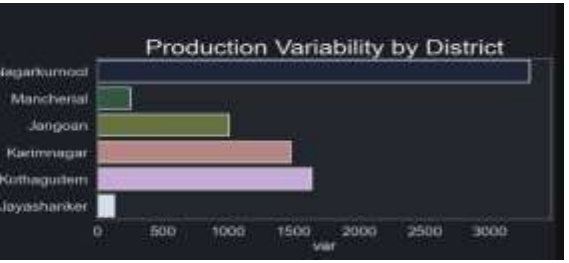






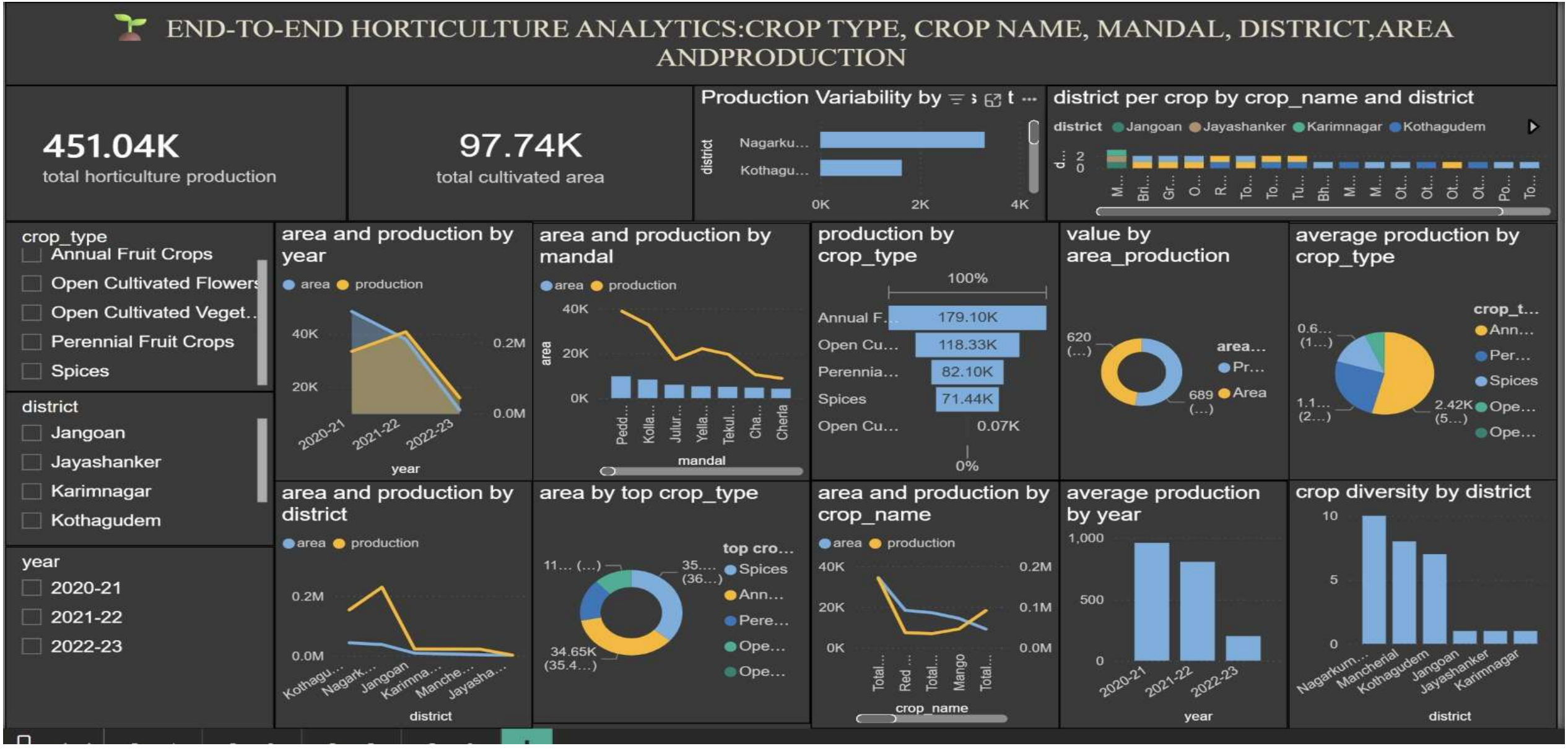






Dashboard : using python

Data visualization: using power bi



Conclusion :

■ Key Metrics

3
Years of Data

Multiple
Districts Covered

Various
Crop Types

■ Tools Used

- Excel (Data Cleaning)
- MySQL (Database)
- Python (Analytics)
- Pandas (DataFrame)
- Matplotlib/Seaborn (Viz)

■ Process Flow

Raw Data → Collection
Cleaned → Preprocessing
SQL DB → Storage
Python → Analysis
Dashboard → Visualization

END-TO-END HORTICULTURE ANALYTICS

Complete Data Pipeline: From Raw Data to Actionable Insights

- Data Collection
- Cleaning
- Storage
- Analysis
- Visualization